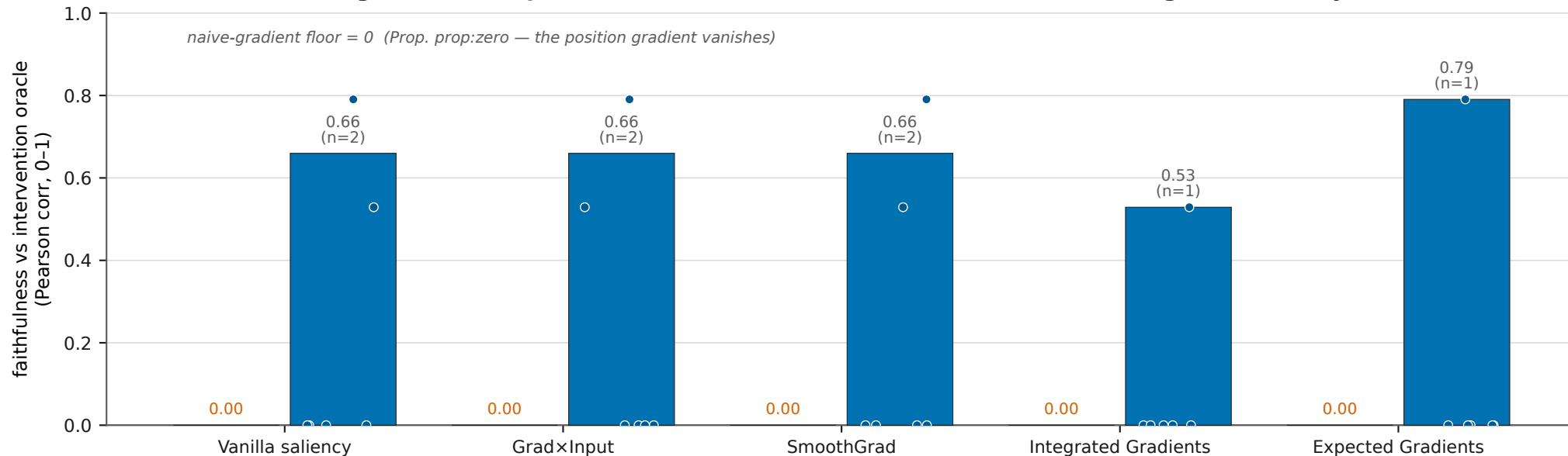


Keystone: the bilinear sampler makes gradient attribution faithful on the position regime — yet recovers no semantics

On the discrete position/index regime the naive gradient is provably zero (Prop. prop:zero), so every gradient method scores faithfulness 0. Turning Paper-1's bilinear index-boundary sampler ON (same surrogate as the SI joystick tool) restores a real position gradient and the methods become FAITHFUL — up to 0.791 (Vanilla saliency on pong). Yet no method names a game concept: semantic recovery is a yes/no question, and the answer is no for all 30 (method × game) records in BOTH conditions. Faithfulness is repairable; the semantic gap is not — the danger zone is not a fixable artifact of the vanishing gradient.

(a) Position/index regime: the sampler RESTORES faithfulness (0 → off the floor) for the gradient family



(b) Semantic recovery is a yes/no question — the answer is “no” for every method, in both conditions (the gap survives the fix)

